Mechanical state inversion method for structural performance evaluation of existing suspension bridges using 3D laser scanning

Abstract: *An abnormal mechanical state will cause structural performance degradation of suspension bridges. This paper presents a mechanical state inversion method for suspension bridges based on the measured intersection points of main cables through 3D laser scanning. First, on the basis of catenary, a mathematical model between suspender force and measured intersection point coordinates is established, and an accurate calculation method for suspender force is proposed. Then, an accurate calculation method for the internal force of suspension bridge components with abnormal suspender force is proposed to evaluate the structural performance. Numerical experiments show that the proposed method is accurate in theory. Lastly, 3D laser scanning is used to capture the shape of the main cable, and an automated algorithm is developed to calculate the intersection point of main cables precisely. The accuracy and reliability of the proposed method are verified on a completed suspension bridge with a span of 808 m.*

1 INTRODUCTION

The frequent collapse accidents of large bridges in recent years have caused a large number of casualties and property losses. In accordance with the statistical data of demolition or collapse in China from 2010 to 2018, the actual average life of these bridges is only 30 years, while the design life is usually more than 50 years. The structures demolished earlier have abnormal response. If these abnormal responses are not evaluated and repaired in time, the bridges will collapse suddenly under external factors, such as the Myaungmya bridge in Myanmar, the Yilan bridge in Taiwan, and the Genoa bridge in Italy. A suspension bridge is a type of long-span structure, whose structural damage may cause disastrous consequences. Therefore, reliable, timely, and comprehensive anomaly detection and evaluation methods are essential for this bridge type.

Intelligent infrastructure (Adeli and Jiang, 2009) has been proposed to solve structural health, detection, and damage diagnosis problems with low cost and high reliability. Many intelligent algorithms and theories have accordingly been proposed in recent years. For example, wavelet (Guan et al., 2016) and Bayesian (Arangio et al., 2012) neural networks have been used to detect the damage of suspension bridges. Numerical simulation results show that these intelligent methods have great potential in theory. However, these methods still need to be proven in practical projects. A suspension bridge is a complex structural system with multiple components. This structure has remarkable mechanical uniqueness. Considerable damage can cause structural dynamic response changes, and a health-monitoring system is deployed on suspension bridges to monitor structural damage (Yong et al., 2013; Koo et al., 2013).

Environment-induced vibration data can be directly used for real-time damage detection of suspension bridges (Siringoringo et al., 2008); Domaneschi et al., 2016). Nevertheless, existing research shows that only the serious damage of the main components, such as the tower and deck, will cause changes in the fundamental frequency and mode of the structure (Domaneschi et al., 2013; Talebinejad et al., 2014; Wang et al., 2015; Koo et al., 2013; An et al., 2019). Different from the stiffness of a girder bridge, that of a suspension bridge is mainly caused by the gravity effect. Local damage will not change the overall stiffness of the structure. Thus, monitoring and evaluating the local damage of suspension bridges (such as suspenders) on the basis of the overall dynamic characteristics remain as challenges.

Several fine detection methods have been proposed to identify component damage (An et al., 2019). These methods mainly focus on the suspenders of suspension bridges. Structural damage usually causes a change in suspender force. The estimation methods for suspender cable force are divided into direct and indirect methods. The direct method is to install strain gauge, optical fiber, pressure ring, and other sensors during cable installation (Li et al., 2011; Ansari et al., 2007; Li et al., 2009; Li et al., 2004). It usually has high measurement accuracy. However, only a few suspenders may be installed with such sensors due to the cost of sensor installation and maintenance. Moreover, damaged sensors cannot be replaced due to the limitation of installation conditions. The indirect method is to estimate the cable force from the fundamental frequency of cable vibration (Zui et al., 1996; Kim et al., 2005). Several days are usually needed to collect and analyze all suspender frequencies manually. Methods based on noncontact vision have also been developed in recent years to solve the problems of data acquisition efficiency and risk (Feng et al., 2017; Kim et al., 2013). Preventing a camera to move for several times to find a suitable shooting position is difficult due to the contradiction between camera resolution and measurement distance. The use of UAV to carry an image system is a more flexible and efficient method (Tian et al., 2020). The color difference between cables and background and the vibration amplitude of cables are the key points of the visual method, which may limit the practical application of this method in many bridges. Cable force estimation based on frequency is the only theoretical method among indirect methods. Frequency methods are influenced by the coupling effect of cable structure, boundary conditions, length, and bulk density (Wang et al., 2017). For each suspender, a model should be modified to determine reasonable parameters. This work is extremely difficult in practical engineering applications. The damage of a suspender may also cause changes in other physical properties, such as the vibration mode (Pandey et al., 1991) and the local magnetic field. These problems require an elaborate detection strategy (An et al., 2015; Christen et al., 2003). Elaborate methods should conduct a detailed detection for each suspender and can only be used in the periodic comprehensive inspection of bridges.

To sum up, although sensor systems have the capability of real-time health monitoring, the anomaly location and quantification are facing challenges. Elaborate methods are effective in local anomaly detection, but they are unsuitable for routine inspection. Suspender force measurement based on a frequency method is also unsuitable for short suspenders in mid-span areas. Existing methods hardly evaluate structural performance changes caused by anomalies. However, structural performance evaluation is an important support for managers to make early warning and maintenance decisions. In this paper, a new method for calculating the suspender force from the intersection point coordinates of main cables is proposed, which solves the problem that frequency methods are inadequately reliable and unsuitable for short suspenders, as shown in Chapter 2. On the basis of the finite element method (FEM), a mechanical state inversion method for existing suspension bridges is proposed, which can accurately calculate the influence of abnormal suspender force on the structural performance, as shown in Chapter 3. In the proposed method, the measured coordinates of the main cable intersection point are the main input parameters. The automation strategy of using 3D laser scanning to measure the intersection point of main cables solves the problems in efficiency, comprehensiveness, and reliability of data acquisition, as indicated in Chapter 4. The experimental results on a large suspension bridge show the effectiveness of the proposed method in practical engineering, as demonstrated in Chapter 5.

2 METHODOLOGICAL FRAMEWORK

FEM, which is usually used in the structural design stage, is mainly adopted to solve for the mechanical state of complex structures. The general flow of FEM is shown in Figure 1(a). However, FEM cannot be directly applied to existing suspension bridge structures due to two main reasons. (1) It is difficult to obtain accurate input parameters, such as the coordinates of main cable nodes, for existing structures with anomalies or damages. (2) The force of hundreds of suspenders cannot be measured accurately, but it is an essential parameter in the calculation of the mechanical state of suspension bridges. The method proposed in this paper overcomes these problems, as shown in Figure 1(b). In detail, the proposed method has five main contents: accurate calculation of main cable intersection, construction of a mathematical model of intersection and suspender force, an inverse modeling method for a finite element (FE) model, force balance judgment of FEM, and a performance evaluation strategy.

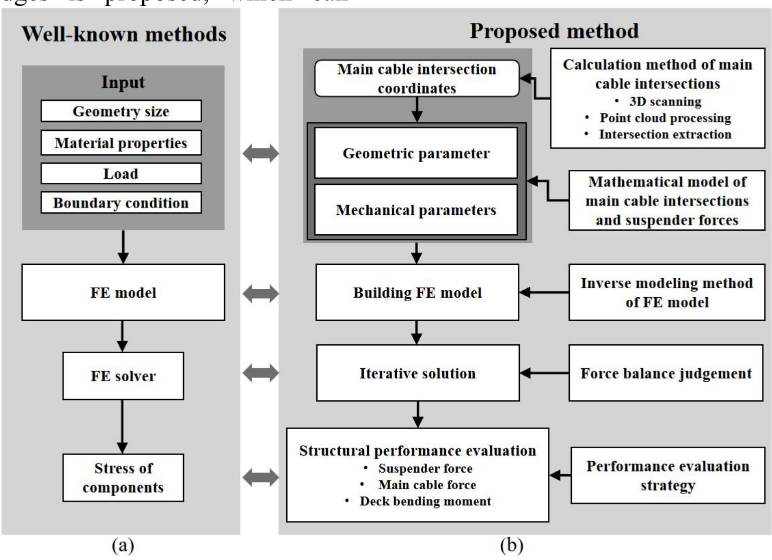

Figure 1 Methodological framework: (a) general flow of FEM, (b) proposed performance evaluation method for suspension bridges.

3 METHOD FOR ESTIMATING SUSPENDER FORCE

3.1 Mathematical model of suspender force

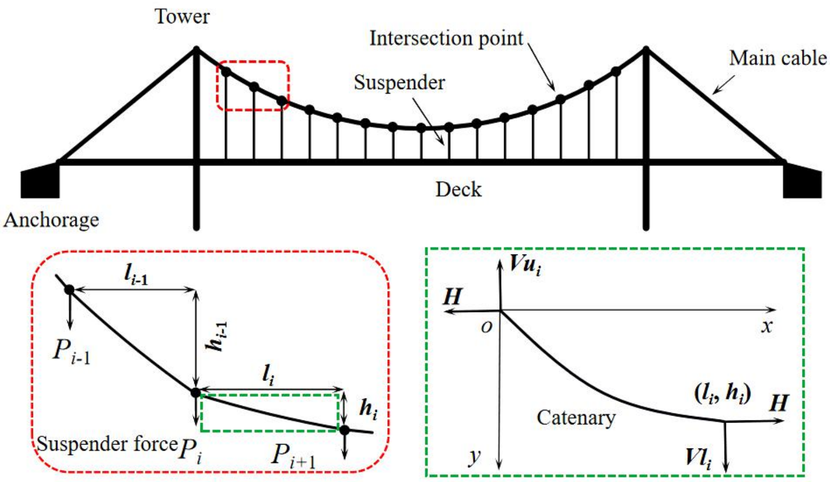

Figure 2 Mechanical analysis diagram of a cable system.

Figure 2 shows the mechanical relationship of a cable system. The bending stiffness of the main cable is assumed to be negligible. Under the action of self weight, the main cable between two adjacent suspenders meets the standard catenary equation. The main cable between ***i*** and ***i+1*** is regarded as the analysis object. The catenary equation of the main cable satisfies the following formulas:

$$\boldsymbol{y} = \frac{\boldsymbol{H}}{\boldsymbol{q}}\left[\cosh\alpha_i - \cosh\left(\frac{2\beta_i \boldsymbol{x}}{\boldsymbol{l}_i} - \alpha_i\right)\right], \text{(1a)}$$

$$\alpha_i = \sinh^{-1}\left[\frac{\beta_i(\boldsymbol{h}_i/\boldsymbol{l}_i)}{\sinh\beta_i}\right] + \beta_i, \text{(1b)}$$

$$\beta_i = \frac{\boldsymbol{q}\boldsymbol{l}_i}{2\boldsymbol{H}}, \text{(1c)}$$

where $\boldsymbol{l}_i$ is the horizontal distance between ***i*** and ***i+1*** intersections, and $\boldsymbol{h}_i$ is the vertical distance. ***q*** is the linear unit weight of the main cable, and ***H*** is the horizontal force in the main cable.

For the derivation of Formula 1a, the inclination angle of any position on the main cable is

$$\tan\theta = -\sinh\left(\frac{2\beta_i \boldsymbol{x}}{\boldsymbol{l}_i} - \alpha_i\right). \text{(2)}$$

Therefore, the vertical force in the main cable at any position can be calculated as

$$\boldsymbol{F} = \boldsymbol{H}\tan\theta. \text{(3)}$$

The vertical component forces $\boldsymbol{Vu}_i$ and $\boldsymbol{Vl}_i$ at the upper and lower ends of the main cable can be obtained by introducing the boundary conditions ***x=0, y=0*** and ***x=li, y=hi*** into Formula 3.

$$\boldsymbol{Vu}_i = \boldsymbol{H}\sinh\alpha_i \quad \text{(4a)}$$

$$\boldsymbol{Vl}_i = -\boldsymbol{H}\sinh(2\beta_i - \alpha_i) \quad \text{(4b)}$$

At ***i*** suspender, the force can be obtained from the balance of vertical force.

$$\sum \boldsymbol{y} = 0 \Rightarrow \boldsymbol{P}_i = \boldsymbol{Vl}_{i-1} - \boldsymbol{Vu}_i \quad \text{(5a)}$$

$$\boldsymbol{P}_i = -\boldsymbol{H}\sinh(2\beta_{i-1} - \alpha_{i-1}) - \boldsymbol{H}\sinh\alpha_i \quad \text{(5b)}$$

In accordance with Formulas 1b, 1c, and 5b, Pi of an arbitrary suspender is related only to $\boldsymbol{l}_i$, $\boldsymbol{h}_i$, $\boldsymbol{l}_{i-1}$, $\boldsymbol{h}_{i-1}$, ***H*** , and ***q***. When an appropriate method is used to measure the main cable intersection coordinates of an existing suspension bridge accurately, the arbitrary suspender force can be calculated using the horizontal force ***H*** and the linear unit weight ***q*** of the main cable. ***q*** can usually be calculated accurately from design drawings. Therefore, the key to calculating suspender force is to determine the horizontal force ***H*** of the main cable of existing suspension bridges.

3.2 Accurate calculation method for *H*

Figure 2 depicts that ***H*** is generated by the joint action of suspender force and the dead weight of the main cable to achieve force balance on the main cable. The suspender force is composed of the dead weights of the suspender and deck. ***H*** caused only by the dead weight of the main cable can be calculated using the catenary formula. ***H*** caused only by the dead weight of the deck can be calculated using a parabola. However, in large suspension bridges, FEM should still be used for the accurate calculation of ***H*** caused by the dead weights of the main cable, suspender, and deck. FEM for existing suspension bridges can be carried out as follows.

3.2.1 Modeling

The cable element is used to simulate the main cable and suspender, and the beam element is used to simulate the deck. The input parameters of FEM include material properties, boundary conditions, geometric parameters, and loads. The material properties are elastic modulus and Poisson’s ratio, which can be obtained from design drawings or test reports. The boundary condition of suspension bridges is clear, that is, the end of the main cable and the tower bottom are consolidated. The geometric parameters are the node coordinates of main cable, suspender, and deck elements. The main cable element and the upper end of the suspender element share the same node, which is the measured intersection point of the main cable. The deck element and the lower end of the suspender element share the same node, which can be obtained from the designed deck curve. Figure 3 shows the FE model of an existing suspension bridge.

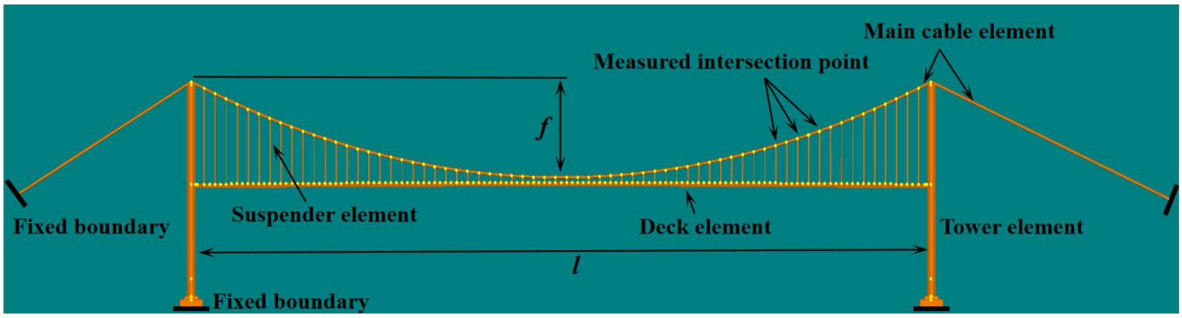

Figure 3 FE model of a suspension bridge.

The geometry of the FE model in Figure 3 is consistent with that of an actual structure with force balance. When self weight is applied to the FE model, the geometric state will deviate from that of the actual structure. This situation is a distinct problem faced by existing suspension bridges in a reverse solution. The proposed method is to input mechanical parameters for offsetting the deformation caused by self weight. The mechanical parameter of the cable system is the unstressed length of the cable element. The relationship between measured geometric and mechanical parameters can be constructed using the following method.

(1) Mechanical parameters of the main cable element

In Figure 1, the microsegment $\boldsymbol{ds}$ is cut out from the catenary (Figure 4), and the catenary length can be obtained by integrating the microsegment $\boldsymbol{ds}$ into the curve. $\boldsymbol{ds}$ can be converted into $\sqrt{1+(\frac{dy}{dx})^2}\,dx$ in Cartesian coordinates. Therefore, the length of the main cable element $\boldsymbol{i}$ in the FE model is

$$\begin{aligned} S_i &= \int_0^{l_i}\sqrt{1+(\frac{dy}{dx})^2}\,dx \\ &= \frac{H}{q}[\sinh(2\beta_i-\alpha_i)+\sinh\alpha_i] \end{aligned} \quad . \text{(6a)}$$

The tension of the main cable is $\boldsymbol{T}$. The extension of $\boldsymbol{ds}$ is $\frac{dT}{E_m A_m}ds$, which is converted into $\frac{H[1+(\frac{dy}{dx})^2]}{E_m A_m}dx$ in Cartesian coordinates. Hence, the extension of the main cable element $\boldsymbol{i}$ is

$$\begin{aligned} \Delta S_i &= \int_0^{l_i}\frac{H[1+(\frac{dy}{dx})^2]}{E_m A_m}dx \\ &= Hl_i[4\beta_i+\sinh(4\beta_i-2\alpha_i)+\sinh(2\alpha_i)]/(8E_m A_m\beta_i) \end{aligned} \quad , \text{(6b)}$$

where $\boldsymbol{E_m}$ is the elastic modulus of the main cable, and $\boldsymbol{A_m}$ is the effective area of the main cable section.

The unstressed length of the main cable element $\boldsymbol{i}$ is

$$S_{0i} = S_i - \Delta S_i \,. \text{(6c)}$$

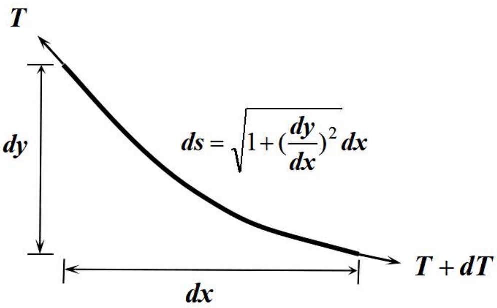

Figure 4 Microsegments of the catenary.

(2) Mechanical parameters of the suspender element

In the FE model, the stressed cable length of the suspender element $\boldsymbol{i}$ is $\boldsymbol{L_i}$. Under the dead weight of the deck, the extension of the suspender is $\frac{P_i-q_sL_i}{E_sA_{si}}L_i$. Under the dead weight of the suspender, the extension of the suspender is $\frac{q_sL_i}{2E_sA_{si}}L_i$. Therefore, the unstressed cable length of the suspender element $\boldsymbol{i}$ is

$$\begin{aligned} L_{0i} &= L_i - \frac{P_i-q_sL_i}{E_sA_{si}}L_i - \frac{q_sL_i}{2E_sA_{si}}L_i \\ &= L_i\left(1-\frac{P_i-(q_sL_i)/2}{E_sA_{si}}\right) \end{aligned} \quad , \text{(7)}$$

where $\boldsymbol{q_s}$ is the linear unit weight of the suspender, and $\boldsymbol{Es}$ is the elastic modulus of the suspender. $\boldsymbol{A_{si}}$ is the area of the suspender section. $\boldsymbol{P_i}$ is obtained using Formula 5b.

Formulas 1b, 1c, 6a, and 6b show that when $\boldsymbol{l_i}$ and $\boldsymbol{h_i}$ have been obtained using the measured intersection coordinates of the main cable, the mechanical parameters of the main cable are related only to $\boldsymbol{H}$. Formulas 5b and 7 show that the mechanical parameters of the suspender are also related only to $\boldsymbol{H}$.

3.2.2 Convergence strategy for force balance

When appropriate $\boldsymbol{H}$ is input, the deformation of the FE model will approach zero under the coupling effect of mechanical parameters, measured geometric parameters, and self weight. This state is considered to have reached force balance, and the FE model results will be consistent with the condition of an actual bridge. The proposed method uses an iterative convergence strategy to estimate $\boldsymbol{H}$ of existing suspension bridges accurately. After any initial value $\boldsymbol{H_0}$ is input, the mechanical parameters of the FE model are calculated using Formulas 6 and 7. The geometry nonlinear solver is used to solve the FE model. When the input value $\boldsymbol{H_0}$ is not equal to the model output value $\boldsymbol{H}$, the input $\boldsymbol{H_0}$ cannot make the structure achieve balance. $\boldsymbol{H_0}$ is replaced with the output value $\boldsymbol{H}$ for iteration. The algorithm for the iterative procedure is shown in Figure 5. The recommended value for convergence condition ζ is 0.001.

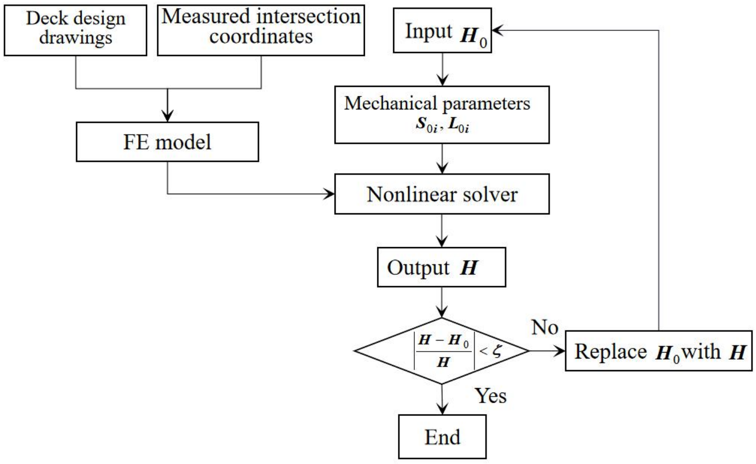

Figure 5 Inversion algorithm for the mechanical state of a cable system.

In an actual structure, the damage of the suspender may cause the redistribution of suspender force in the local area. The area reduction of the suspender section caused by damage is difficult to determine in advance. When the mechanical parameters of the suspender are calculated, the design section area is still used, which may cause calculation deviation of the tension of the damaged suspender. However, given that the dead weight of the deck will not change, the damage of the suspender will not evidently change the overall distribution of suspender force on the main cable. Therefore, calculating $\boldsymbol{H}$ by using the iterative method in Figure 5 is accurate.

The initial design horizontal force can be used as the initial value for iteration. When the design value of $\boldsymbol{H}$ is missing, the approximate value of $\boldsymbol{H}$ can be estimated using the parabola method as the initial value $\boldsymbol{H_0}$.

$$\boldsymbol{H}_0 = \frac{(\boldsymbol{q}+\boldsymbol{q_D})\boldsymbol{l}^2}{8\boldsymbol{f}}, \quad (8)$$

where $\boldsymbol{q_D}$ is the line unit weight of the deck. $\boldsymbol{l}$ is the bridge span, and $\boldsymbol{f}$ is the relative height of the main cable, as shown in Figure 3.

4 PERFORMANCE EVALUATION OF AN EXISTING SUSPENSION BRIDGE

4.1 Performance evaluation method

In accordance with the method proposed in Chapter 3, when the intersection coordinates of main cables are obtained, the mechanical state of suspension bridges can be evaluated. The detailed evaluation strategy can be carried out on the basis of the method shown in Figure 6. The damage or abnormality of the suspender usually causes a change in suspender force. When $\boldsymbol{H}$ is known, the abnormal suspender force can be detected rapidly by measuring the intersection coordinates. Once the abnormal suspender force is detected, the influence of the change in suspender force on the internal force of the deck and other components is evaluated accurately.

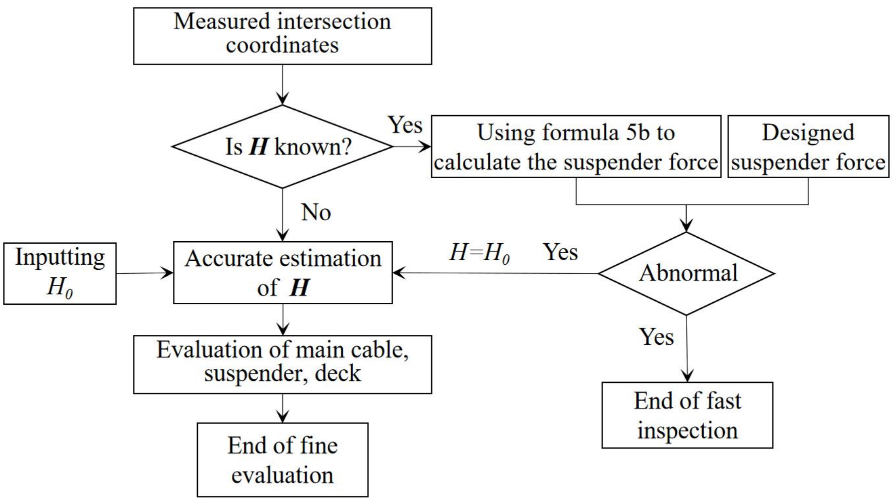

Figure 6 Performance evaluation process

When the actual internal force (axial force and bending moment) of the main cable, suspender, and deck exceeds the allowable value, serious accidents will be caused. Therefore, the actual internal force of each component is the main index to reflect the structural performance.

When the suspender is not damaged, the calculation results of the FE model obtained using the method in Figure 5 are consistent with the internal forces of each component of an actual bridge. When the suspender is damaged, the internal force of the main cable, suspender, and deck can be accurately calculated in accordance with the following method.

(1) Main cable

Generally, the suspender is strictly vertical, and the horizontal force $\boldsymbol{H}$ in the main cable is equal everywhere. When $\boldsymbol{H}$ is obtained, the tension of any main cable segment can be calculated as

$$\boldsymbol{T} = \boldsymbol{H} / \cos\boldsymbol{\theta} , (9)$$

where $\boldsymbol{\theta}$ is calculated using Formula 2.

(2) Suspender

$\boldsymbol{H}$ value obtained by iterative convergence is integrated into Formulas 5, and the tension of each suspender is calculated from the measured intersection coordinates. Regardless of whether the suspender is damaged or not, this method is accurate.

(3) Deck

When all suspender forces are known, the deck can be isolated from the FE model of the suspension bridge and analyzed separately, as shown in Figure 7. The suspender elements are replaced with upward node forces. The node force value is equal to the corresponding suspender force value. The suspender force obtained from Formula 5 includes the dead weights of the suspender and cable clamp. The weights of the suspender and cable clamp should be deducted from the node force.

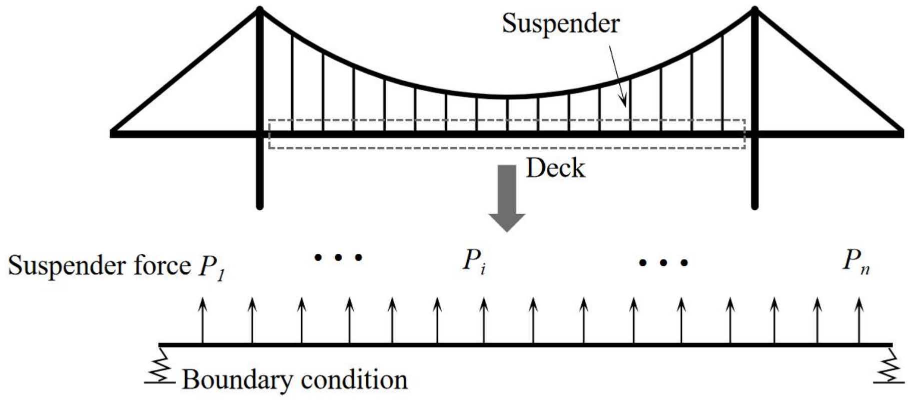

Figure 7 Internal force calculation model of the deck

4.2 Numerical experiment verification

Numerical experiments are performed to prove that the proposed method is correct in theory. In a FE model of a suspension bridge with a span of 800 m, artificial damage is used to simulate the change in structural mechanical state. In an actual structure, an abnormal suspender force is usually caused by the suspender damage. For example, some or all steel wires of a suspender are broken due to fatigue. Therefore, in the FE model, the suspender damage is simulated by reducing the section area of the suspender element. In the numerical experiments, different damage degrees (25%, 50%, and 75%) are set for the damaged suspender to verify whether the proposed method is adequately sensitive before the suspender is completely broken. In suspension bridges, the length of suspenders at different positions varies. The changes in internal force caused by the damage of long and short suspenders are different. Thus, in the numerical experiment, the 3rd, 17th, and 33rd suspenders are selected as the damaged suspenders. The damage details are shown in Table 1. The FE model with damage is analyzed accurately. This model is a forward solving process. The proposed method is a reverse process. This numerical experiment does not consider the geometric measurement errors that may exist in practice to prove that the proposed method is accurate in theory. Therefore, the coordinates of the main cable intersection obtained by forward analysis are regarded as the measured coordinates of the proposed method. The initial value $\boldsymbol{H_0}$ of the main cable is calculated using a parabola method.

Table 1

Detailed damage parameters of suspender

Suspender number	*Suspender position*	*Length*	*Damage degree*
3rd	Close to tower	71.5 m	25%, 50%, 75%
17th	L/4	23.7 m	25%, 50%, 75%
33rd	Middle span	4.1 m	25%, 50%, 75%

The proposed method has a fast convergence speed. After four iterations, $\boldsymbol{H}$ achieves the convergence condition, as shown in Figure 8(a). The three damages causing the change in suspender force are accurately reflected in the change in main cable intersection coordinates, as shown in Figure 8(b). The suspender force calculated using the proposed method is consistent with that calculated using the FE model, as shown in Figure 8(c). For the change in deck bending moment caused by suspender damage, the proposed method is also consistent with the FE model, as shown in Figure 8(d). The experimental results indicate that solving the mechanical state of suspension bridges by using the intersection coordinates of main cables is accurate and reliable in theory. Furthermore, the calculation results show that for the long suspender (3rd), 25% damage causes 12.8% of the suspender force change, and the change in the main cable intersection coordinates is up to 10 mm. For the short suspender (33rd), the coordinate change of the main cable intersection caused by the change in 4.4% of suspender force is 3.5 mm. Therefore, the proposed method is sufficiently sensitive to detect the abnormal suspender force of suspension bridges.

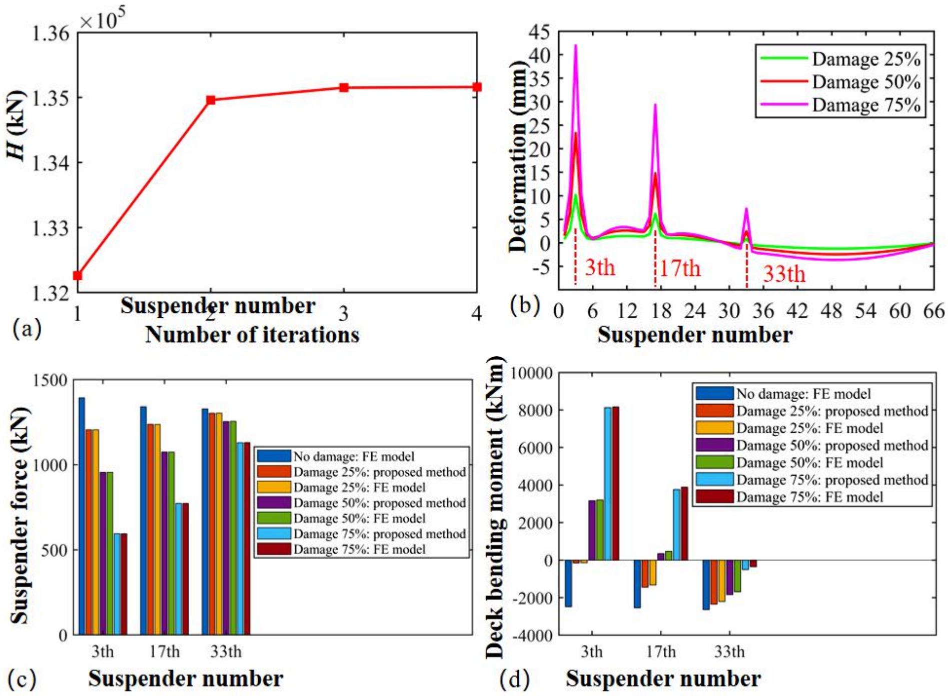

Figure 8 Numerical experiment results: (a) convergence speed of ***H***, (b) deformation of the main cable caused by the suspender damage, (c) cable force of the damaged suspender, (d) bending moment of the deck in the damage position.

5 AUTOMATIC MEASUREMENT METHOD FOR THE MAIN CABLE INTERSECTION

Potential methods for measuring the coordinates of intersection points are (1) installing position sensors (prism, DGPS, etc.) and (2) noncontact geometric measurement methods (Park et al., 2007). For abnormal inspection of all suspenders or for performance evaluation of bridges, the intersection coordinates of all main cables should be obtained. In existing health-monitoring systems, only a few sensors are installed on the deck and tower. The costs for installing and maintaining hundreds of sensors are high. In this paper, a method based on 3D laser scanning is proposed to measure the intersection coordinates of main cables. The method includes four steps: data acquisition, point cloud registration, point cloud segmentation, and intersection extraction.

5.1 Data acquisition

Leica P50 is used as the data acquisition equipment in this study given the relatively high accuracy of a terrestrial laser scanner. It achieves an excellent balance in measuring range, accuracy, and efficiency. The performance parameters of a Leica P50 scanner are summarized in accordance with the test results of an actual bridge, as shown in Table 2. The geometry of a suspension bridge is sensitive to temperature change. Therefore, the time spent in a single scan is a key parameter. For suspension bridges with spans less than 500 m, scanning modes 7 and 8 are recommended. For suspension bridges with spans greater than 500 m, scanning modes 11, 12, and 15 are recommended. The suggested scanning modes achieve a balance in resolution and efficiency in practical engineering, which makes the main cable and suspender point cloud have high point density, and the time required is sufficiently short.

Table 2
Parameters of Leica P50 scanner

Scanning mode	*Parameter*			*Applicable object*
	Maximum range (m)	*Resolution (mm)*	*Time (min)*	
1	120	0.8	54	Steel bars, bolts, etc.
2	120	1.6	13	Stay cable, suspender, etc.
3	120	3.1	3	Suspender, main cable, etc

4	120	6.3	2	Beam, tower, arch and main cable, etc
5	270	0.8	108	Steel bars, bolts, etc.
6	270	1.6	27	Stay cable, suspender, etc.
7	270	3.1	6	Suspender, main cable, etc
8	270	6.3	2	Beam, tower, arch and main cable, etc
9	570	0.8	217	Steel bars, bolts, etc.
10	570	1.6	54	Stay cable, suspender, etc.
11	570	3.1	13	Suspender, main cable, etc
12	570	6.3	3	Beam, tower, arch and main cable, etc
13	>1000	1.6	109	Stay cable, suspender, etc.
14	>1000	3.1	27	Suspender, main cable, etc
15	>1000	6.3	6	Beam, tower, arch and main cable, etc

Resolution is the point spacing of 10 meters from the scanner.

Figure 9 shows the point cloud model of a suspension bridge collected using scanning mode 12. In this case, the main cable near the tower top is 90 m above the deck. The point cloud of the main cable still has high point density. This condition ensures the accuracy of calculating the coordinates of the main cable intersection point on the basis of point clouds.

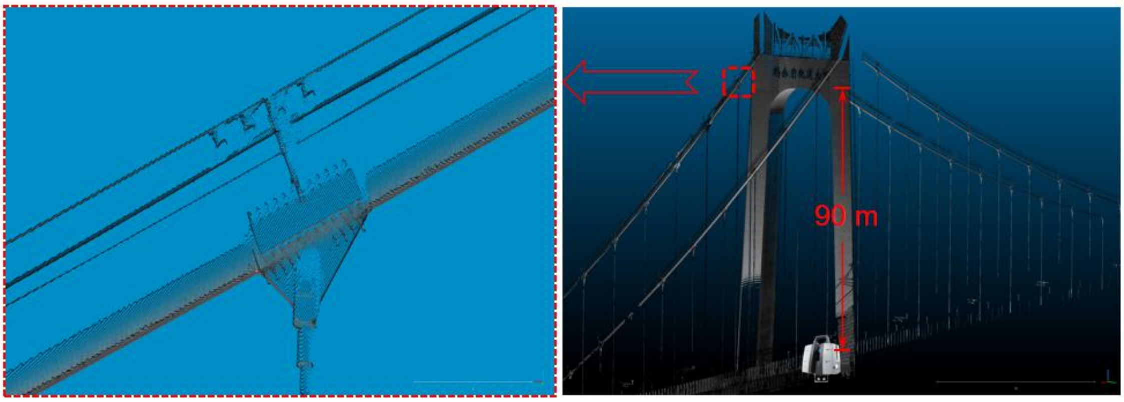

Figure 9 Point cloud of the main cable obtained using Leica P50.

5.2 Accurate registration method for point clouds

Many existing studies propose feasible methods for point cloud registration (Han et al., 2013). One widely used method is ICP (Chen et al., 1991; Besl et al., 1992). The principle of ICP is to find the closest point pair in two point clouds and use the distance matrix of the nearest point pair to estimate the rotation and translation parameters of the moved point cloud. However, the estimated transformation parameters have errors because the minimum distance of point pairs is not equivalent to accurate registration. Therefore, ICP is mainly suitable for dense point clouds with a small geometric size, such as mechanical structure parts. For a suspension bridge point cloud with a large geometric size, the error of the rotation parameters obtained by ICP will cause a registration error that cannot be ignored at several hundred meters. The tower and deck of a suspension bridge usually contain a large number of plane shapes. The proposed method is to use plane features for registration. The specific algorithm steps are as follows:

Step 1: coarse registration. The proposed accurate registration method assumes the deviation after coarse registration to be as small as possible. Thus, the proposed coarse registration method is to use Leica Cyclone for manual coarse registration. The original point cloud is shown in Figure 10(a). The effect of coarse registration is shown in Figure 10(b).

Step 2: finding planar features. After coarse registration of point clouds A and B, the common parts of A and B are divided into blocks by using a cube box. Least square plane fitting is performed for each point cloud block of A and B. The standard deviation of fitting is used to judge whether the point cloud blocks meet the plane characteristics. The plane features of the common areas of A and B are obtained by traversing all point cloud blocks, as shown in Figure 10(c).

Step 3: generation of accurate registration point pairs. On the plane feature of point cloud A, point set {A} with uniform spacing is generated randomly. Point set {A} is projected onto the plane feature of point cloud B, and point set {B} is obtained. Point sets {A} and {B} constitute the registration point pairs in the ICP algorithm, as shown in Figure 10(d).

Step 4: iterative registration. The transformation matrix $\boldsymbol{T}$ from point set {B} to point set {A} is calculated using the singular value decomposition method. Point cloud B is updated with transformation matrix $\boldsymbol{T}$.

Step 5: repeating steps 2–4. The iterative convergence condition of the ICP algorithm is satisfied.

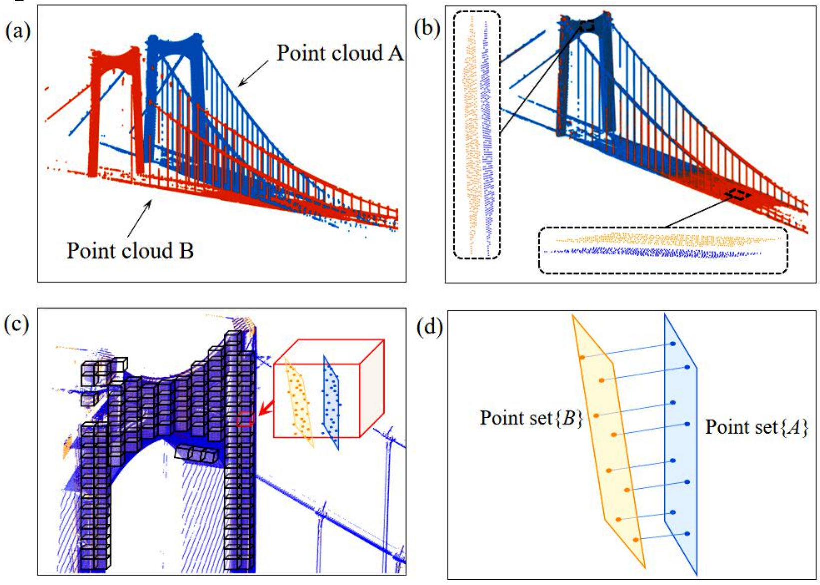

Figure 10 Accurate registration strategy for a suspension bridge point cloud: (a) before coarse registration, (b) after coarse registration, (c) finding plane features, (d) generating accurate registration point pairs.

The accurate registration algorithm takes advantage of the fact that the tower and deck of a suspension bridge contain a large number of plane features and improves the registration point pair generation method in the ICP algorithm. On the basis of this method, the plane features of point cloud A and point B will exactly coincide after iteration.

5.3 Segmentation method for main cable point clouds

Suspension bridges may have hundreds of suspenders. An automatic program should be developed to extract intersection points from point clouds. Only the segment between two suspenders on the main cable of a suspension bridge can meet the catenary mechanical model in Figure 1. Therefore, the target of automatic point cloud segmentation is the main cable segment between the suspenders. Compared with the geometric characteristics of industrial buildings, those of a bridge structure have particularity. Usually, the cable system and tower are in the vertical plane, and the deck is in the horizontal plane. The proposed segmentation method is to count 3D point clouds in a low dimension. In Euclidean space, the point density per unit scale is defined as

$$\rho = \frac{\boldsymbol{Num}}{\Delta E}, \quad (10)$$

where ΔE is the unit scale. In 1D space, ΔE is the unit length. In 2D space, ΔE is the unit area. In 3D space, ΔE is the unit volume. $\boldsymbol{Num}$ is the number of points in ΔE.

Formula 10 is used to calculate the density of the point cloud of the suspension bridge in different dimensions, and the point density parameters of diverse components can be obtained. Figure 11(b) is the 2D point density distribution of Figure 11(a) on the XY plane. Figure 11(c) is the 1D point density distribution in the Y direction. Figure 11(d) is the 1D point density distribution in the X direction. The two density peaks in Figure 11(c) are the locations of the two cable planes of the suspension bridge. After this substep, the main cable and suspender point cloud are separated. The position of each suspender in the span direction can be determined using the peak value of point density in Figure 11(d). The suspender is connected with the main cable by using a cable clamp, the length of which is usually less than 2 m. Hence, the range of 1 m beyond the suspender position is considered the standard segment of the main cable.

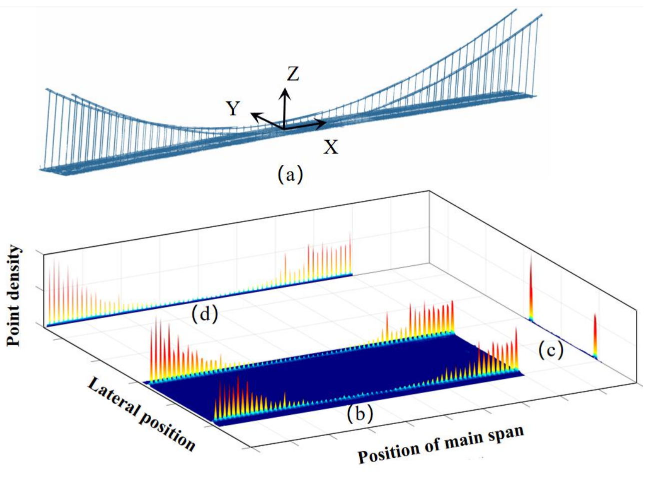

Figure 11 Point density distribution of suspension bridge point clouds: (a) overall point cloud after accurate registration,

(b) point density distribution in the XY plane, (c) point density distribution in the Y direction, (d) point density distribution in the X direction.

As shown in Figure 12, protective devices are usually installed on the main cable. The proposed method for separating the main cable point clouds is as follows:

(1) The local coordinate system is established on the plane perpendicular to the main cable axis. Segment point cloud is projected to the xy plane of the local coordinate system. This step transforms the recognition of a cylinder in a 3D point cloud into circle recognition in a plane point cloud.

(2) Three points are randomly sampled from the plane point cloud. The coordinates ($\boldsymbol{x_i}$, $\boldsymbol{y_i}$) and radius $\boldsymbol{R_i}$ of the circle composed of the three points are calculated. This step is repeated $\boldsymbol{n}$ times to obtain the radius sample $\{\boldsymbol{R_i}\}$.

(3) In accordance with the probability density statistics of $\{\boldsymbol{R_i}\}$, the density peak is regarded as the radius $\boldsymbol{R_0}$ of the circle feature in the plane point cloud. The center coordinates ($\boldsymbol{x_0}$, y_0) corresponding to $\boldsymbol{R_0}$ are regarded as the center of the circle feature in the plane point cloud. In the projection plane, the points within the range of $(x - x_0)^2 + (y - y_0)^2 < (R_0 + \delta)^2$ are regarded as the point clouds of the main cable. The value of δ is related to the minimum distance between the protective devices and the main cable surface.

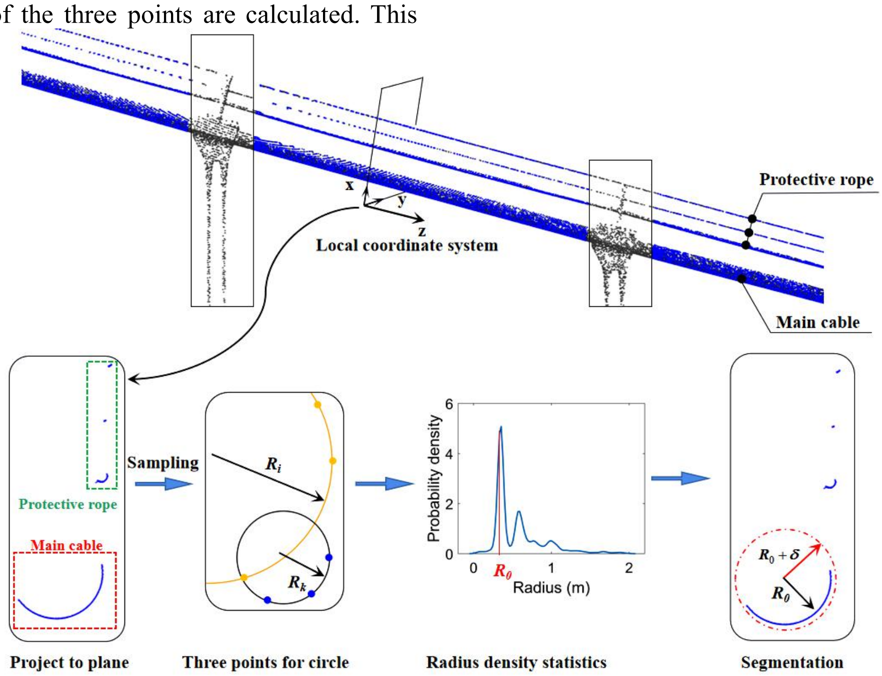

Figure 12 Segmentation process of the main cable segment point cloud.

5.4 Calculation method for the intersection point

The intersection point can be solved using the centerline equations of two adjacent main cable segments. However, the estimation of the parameters of the catenary equation from a point cloud is a nonlinear regression model that cannot be applied using the direct method. This paper suggests a simple and efficient method. A section of the main cable point cloud is intercepted on both sides of the suspender. Centerlines 1 and 2 of the cylinders are obtained using the cylinder-fitting method (Abdul et al., 2019), as shown in Figure 7. No intersection exists between two 3D lines because of the cylinder fitting error. The two centerlines are projected into the XZ plane. The elevation points of centerlines 1 and 2 at the suspender position $\boldsymbol{x_s}$ are the intersection of the main cable and suspender, as shown in Figure 13. The intersection of center 1 and centerline 2 at the suspender position may not coincide due to the existence of fitting error. The mean value of the two intersections is recommended to use as the main cable intersection point.

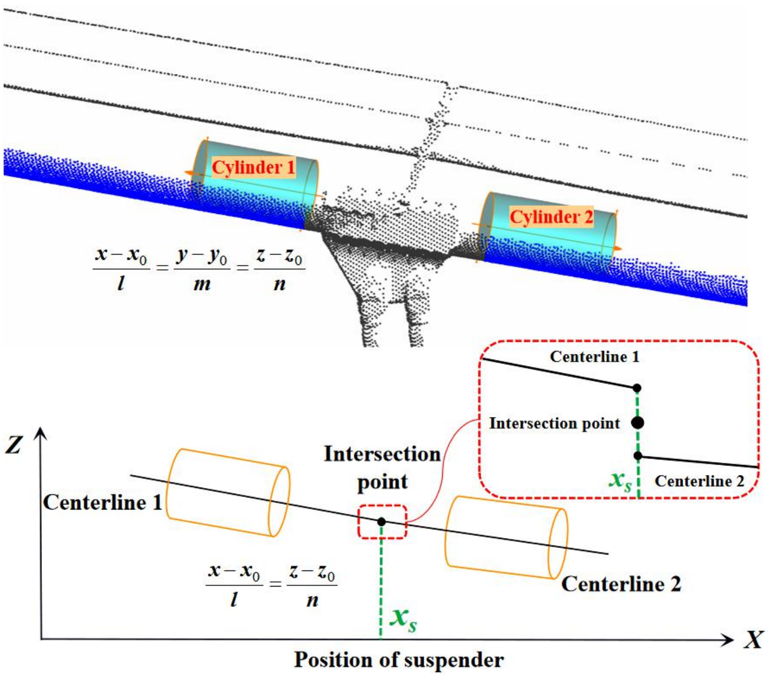

Figure 13 Calculation method for the main cable intersection point.

6 REAL BRIDGE VERIFICATION

Different from the stiffness of an ordinary structure, that of a suspension bridge is mainly produced by gravity, rather than the section size of components. Consequently, a scale experiment, although its test data may be better, cannot prove the effectiveness of the proposed method in practical application. A long-span suspension bridge in Chongqing, China is selected as the experimental object.

6.1 Introduction of the experimental bridge

The Taihong Yangtze River Bridge is a ground-anchored suspension bridge with a main span of 808 m and 132 suspenders. The construction procedure of the ground-anchored suspension bridge is as follows: (1) the installation of the main cable is completed; (2) the deck segment is hung on the main cable with hinge joint (Figure 14(a)); (3) after the installation of all decks, the deck segments are welded as a whole. On the basis of this construction method, the suspender force is exactly equal to the weight of the deck segment at the time of bridge completion. During the service period, when the suspender is damaged, the suspender force and the internal force of the deck will change.

The construction of the Taihong Bridge has just been completed and has not entered the service stage. Figure 14(b) is a real picture of the completion of the Taihong Bridge. The actual suspender force of the Taihong Bridge can be considered accurate as the weight of deck segments, which can be used as the theoretical value compared with the experimental results. Therefore, using the Taihong Bridge as the experimental object to verify the reliability of the proposed method is reasonable.

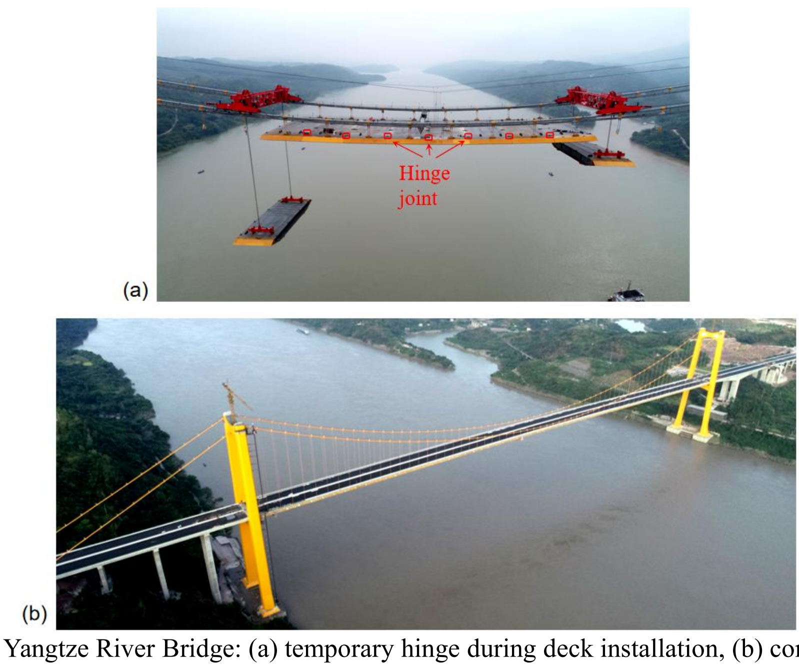

Figure 14 Taihong Yangtze River Bridge: (a) temporary hinge during deck installation, (b) completed state.

6.2 Experimental content

The verification content of the real bridge experiment includes two parts: (1) the reliability of using 3D scanning to measure suspender force and (2) the reliability of the mechanical state inversion method for suspension bridges. Under no-load condition (Figure 15(a)), the absolute accuracy of the proposed method is verified by comparing the estimated suspender force with the weight of each beam segment. Making suspender damage artificially in an actual bridge is difficult and dangerous. To simulate the change in structural mechanical state caused by an abnormal suspender force, a truck load layout scheme is designed to act on the Taihong Bridge. Figure 15(b) shows the scene of scanning after implementing truck load. A total of 36 trucks with a dead weight of 350 kN are used. The truck layout is shown in Figure 15(c). The relative accuracy of the proposed method is verified using the suspender force increment caused by the trucks. The reliability of the proposed mechanical state inversion method is verified by measuring the change in deck internal force caused by the trucks.

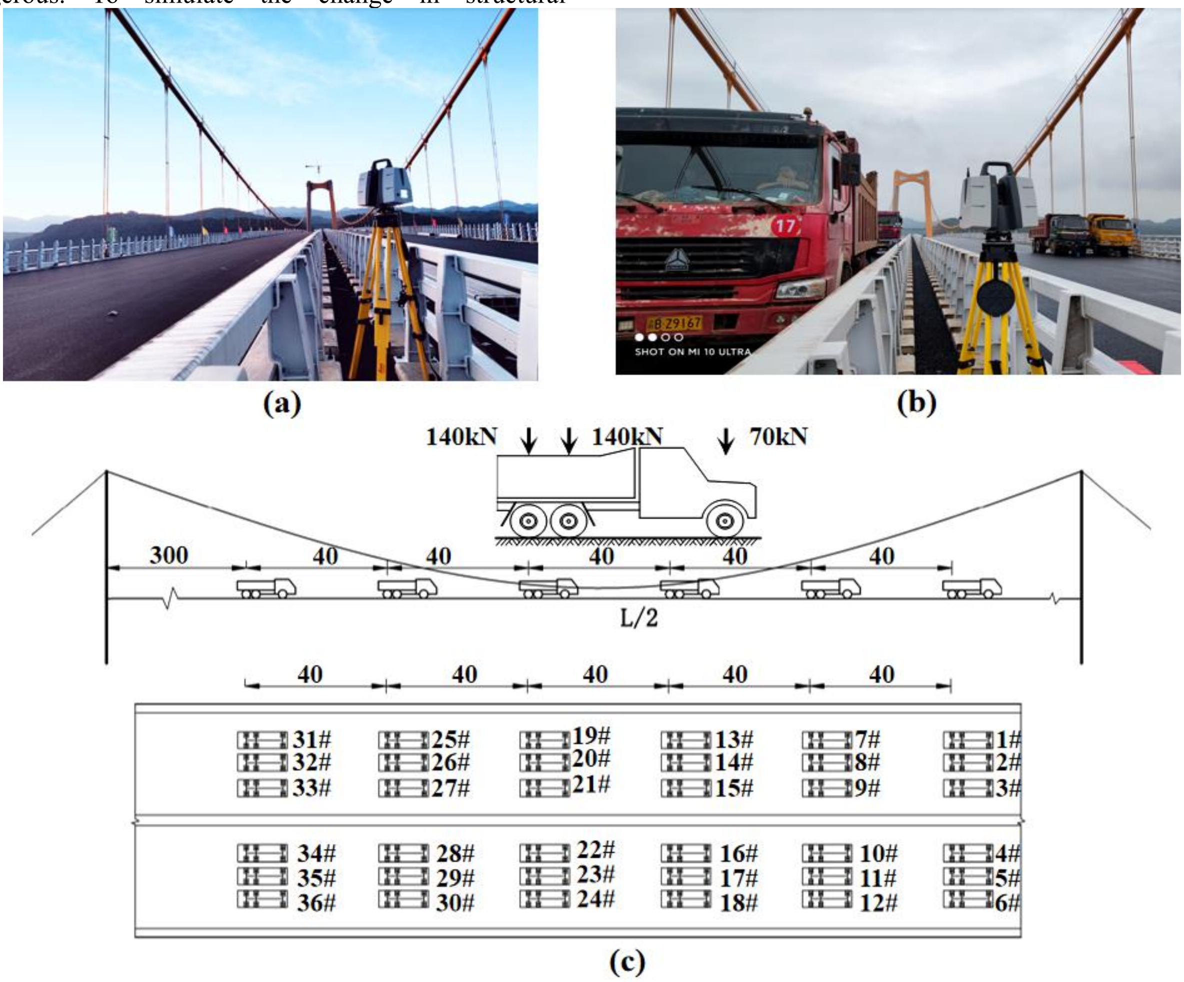

Figure 15 Experimental scheme of the Taihong Bridge: (a) no-load state, (b) loaded state, (c) layout scheme of trucks.

Scanning mode 12 is used to scan the Taihong Bridge. Seven scanning stations are established on the main and side spans. The scanner installation and data acquisition take approximately 60 min. The point cloud model of the Taihong Bridge after accurate registration is shown in Figure 16.

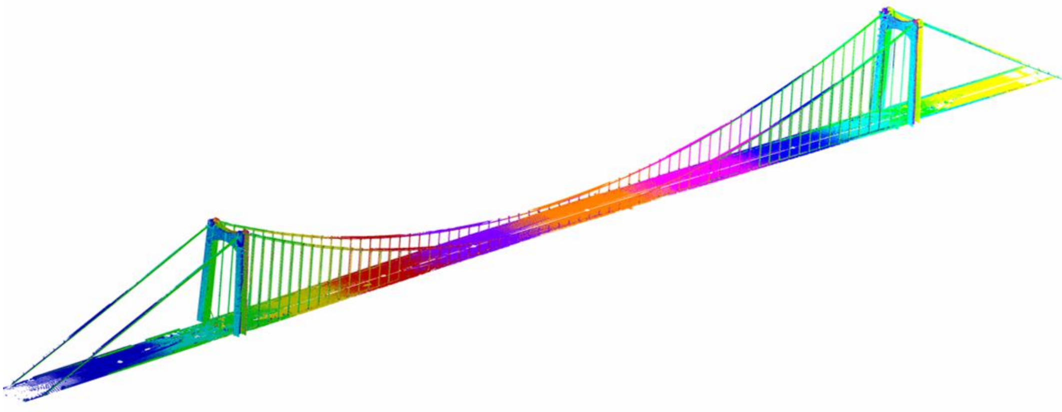

Figure 16 Scanning point cloud model of the Taihong Bridge.

6.3 Experimental results of suspender force

Under no-load condition, the test results of suspender force by using the proposed method are shown in Figure 17(a). The suspender force P_i calculated using the proposed method is the sum of the weights of the deck segment, cable clamp, and suspender. After the actual weights of the deck segment, suspender, and cable clamp of the Taihong Bridge are added, the statistical results of percentage deviation from the measured value are shown in Figure 17(b). The maximum error of the measured value is 8%. The mean value of error percentage is 0.1%, and the standard deviation is 3.4%. The comparison results show that the proposed method has high accuracy and reliability for long and short suspenders. Given that the proposed method is accurate in theory, the main source of error is the coordinate measurement of the main cable intersection. To analyze the accuracy of 3D laser scanning in measuring intersection coordinates, the intersection deformation $\boldsymbol{k}$ is defined, as shown in Figure 17(c). When no suspender force exists at point $\boldsymbol{i}$, the curve between points $\boldsymbol{i-1}$ and $\boldsymbol{i+1}$ is a standard catenary (the dotted line in Figure 17(c)). The suspender force $\boldsymbol{Pi}$ will cause the deformation $\boldsymbol{k_i}$ of point $\boldsymbol{i}$. Figure 17(d) shows the intersection deformation $\boldsymbol{k}$ calculated from the measured intersection coordinates and the theoretical suspender force. The maximum deviation between the measured and theoretical values is 4.5 mm, and the standard deviation is 2.1 mm. The average value of intersection deformation $\boldsymbol{k}$ caused by suspender force is 71.6 mm. The proposed method using 3D laser scanning to measure intersection points is proven to have high geometric accuracy.

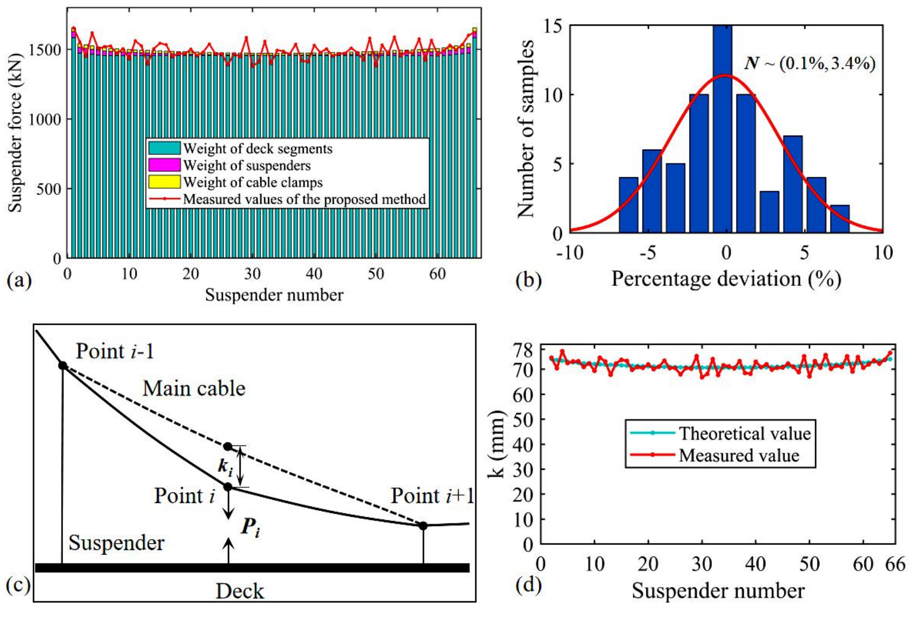

Figure 17 Experimental results of suspender force without load: (a) measured suspender force, (b) error analysis of suspender force measurement, (c) diagram of intersection deformation, (d) measured and theoretical values of intersection deformation.

The designed FE model is used to calculate the actual suspender force of the Taihong Bridge under truck loads. The designed truck loads substantially change the suspender forces in the mid-span region. The suspender force in the mid-span area is increased by 21.4%, as shown in Figure 18(a). The measured value of suspender force by using the proposed method is in good agreement with the theoretical calculation value. The standard deviation of cable force error percentage is 3.5%, which is close to that without truck loads. The two test results (264 samples) before and after loading show that the proposed method has high robustness.

Furthermore, the cable force increment caused by trucks is compared with the theoretical calculation increment. The standard deviation of the percentage error of cable force increment is 1.8%, and the maximum error is 4%, as shown in Figure 18(b). The measurement error of cable force increment is significantly less than the absolute error of cable force. The numerical experiments in Chapter 4.2 have proven that the proposed method is accurate in theory. Therefore, the measurement error of suspender force is generated in the process of using 3D scanning to estimate the intersection coordinates. In the process of fitting a cylinder on the basis of point clouds, a calculation error of centerline exists, which is a random error. The actual cross section of the main cable also deviates from the standard circle, which can be regarded as a system error. When fitting cylinders in two periods of point clouds, the selected position of main cable is the same because the same point cloud-processing method is used. Consequently, the system error is eliminated in the subtraction process. Therefore, given the same point cloud-processing method in the periodic detection of suspension bridges, the proposed method has a resolution of less than 2% of the cable force variation.

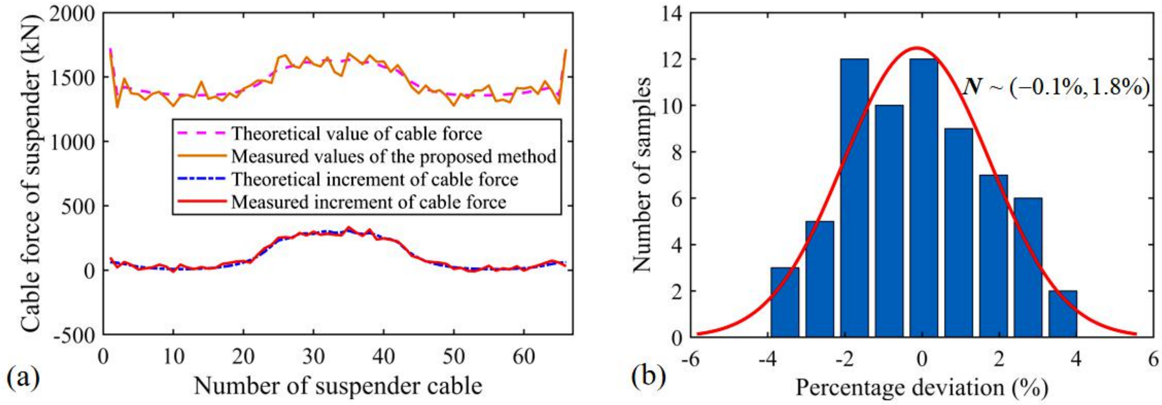

Figure 18 Test results of suspender force under truck load.

6.4 Mechanical state test results

The designed truck layout changes the distribution of suspender force. Under the coupling action of suspender force, dead weight, and 36 trucks, the internal force state of the deck also changes considerably. In accordance with the method provided in Chapter 3.1, the deck in the FE model of the Taihong Bridge is isolated and solved separately. The suspender element is replaced with the measured suspender force through 3D scanning. The actual deck internal force changes caused by 36 trucks are measured by installing strain sensors on the deck. Figure 19 shows the layout of strain sensors and the field data acquisition.

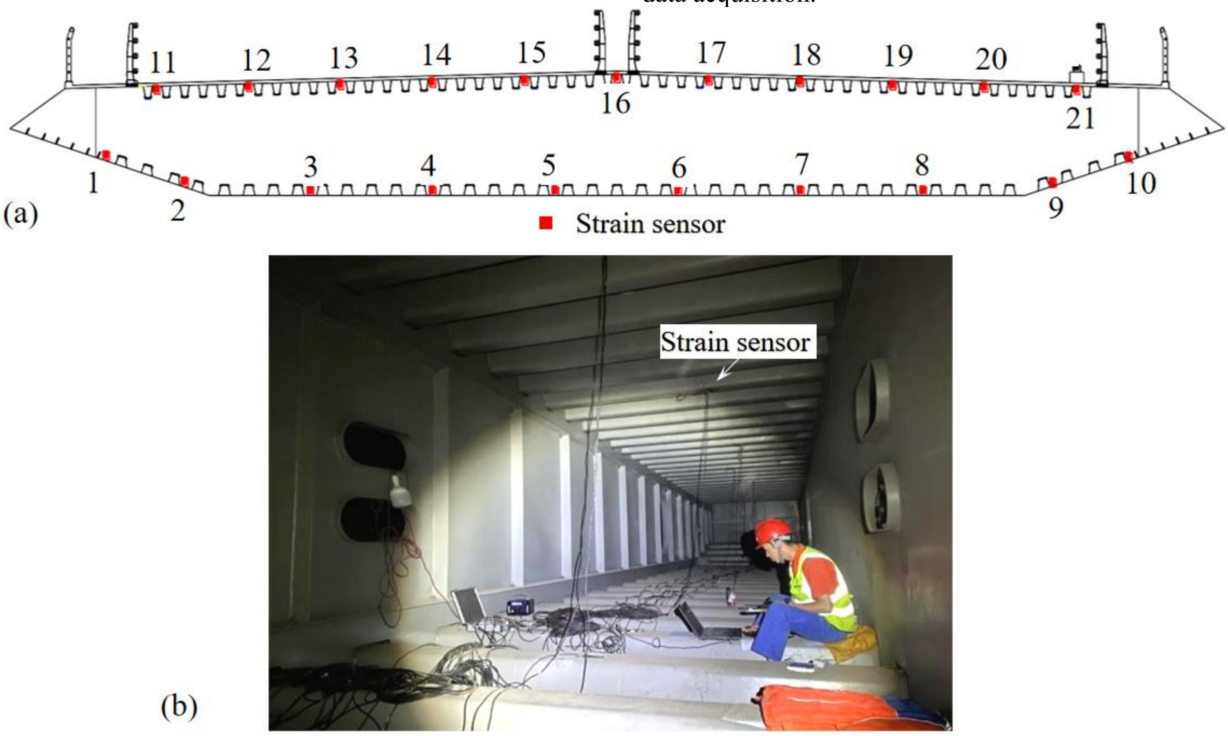

Figure 19 Strain measurement: (1) installation position of strain sensors, (b) strain measurement.

The bending moment at any position of the deck can be obtained using the proposed method, as shown in Figure 20(a). The deck section shown in Figure 19(a) is located at L/2 of the main span. In the truck area, substantial bending moments are generated on the deck. Figure 20(b) shows the strain calculated using the proposed method and the strain measured using sensors. The comparison results show that the bending moment of the deck and the stress (strain) at any position in the deck section can be calculated accurately by using the proposed method.

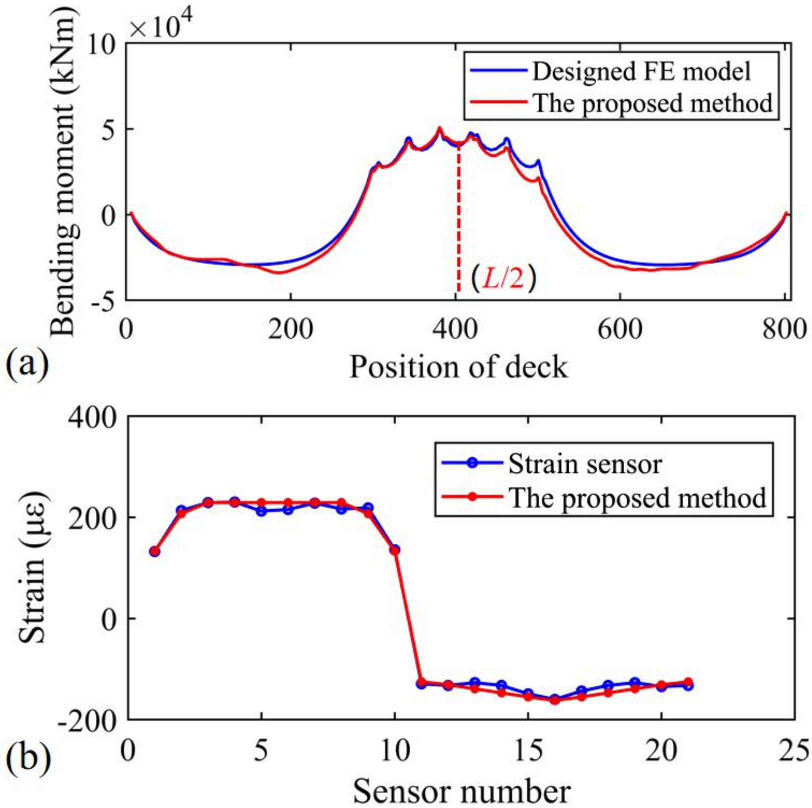

Figure 20 Mechanical state of the deck: (a) deck bending moment calculated using the proposed method, (b) measured and calculated strain.

7 CONCLUSION

In this paper, a method for the mechanical state inversion of suspension bridges based on the measured intersection of main cables is proposed. The main contributions of this work are as follows: (1) a new computational theory for calculating the suspender force of a suspension bridge on the basis of the intersection point of the main cable is established; (2) an automatic strategy, which uses 3D laser scanning to measure the coordinates of the main cable intersection point efficiently and accurately, is proposed; (3) a reverse calculation method for the mechanical state of a suspension bridge is proposed. The specific conclusions are as follows:

(1) Compared with the local deformation of a deck, the local deformation of the main cable caused by an abnormal suspender force is greater. The abnormality of a suspender is easy to locate and quantify accurately by monitoring the change in the intersection point of the main cable. This study can provide a new direction for the health monitoring of suspension bridges.

(2) The accuracy of the established cable force calculation theory is related only to the coordinate measurement error of the main cable intersection point. Compared with the frequency method, the proposed method has stronger adaptability and reliability. It is a new theoretical method for the accurate measurement of the suspender cable force of a suspension bridge.

(3) The proposed method can accurately estimate the internal force changes caused by an abnormal suspender force. It is a reverse method to calculate the internal force of a structure on the basis of the measured geometry of the existing structure. The accuracy of the method is proven via the load experiment of a real bridge.

(4) The experimental results of a real bridge show that 3D laser scanning has remarkable accuracy advantages and data reliability. The proposed automatic strategy for point cloud registration, segmentation, and intersection calculation reduces the time spent on the comprehensive detection of large suspension bridges to less than a few hours. Thus, fine detection and performance evaluation can be conducted in routine inspection, which cannot be completed in previous methods.

Although real bridge experiments have verified the capability of the proposed method to be applied in practical engineering, further quantitative analysis of the error of measuring intersection coordinates using 3D scanning is needed, so that the proposed method can detect more minor damage.

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